Talk it through: inertia

Language goal: I can **explain** why things keep moving.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · A PUCK SLIDES ON ICE AFTER THE PUSH



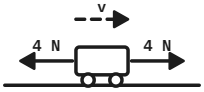
no push · keeps going

A ASKS "Why does the puck keep moving after the stick lets go?"

B SAYS "The net force is about ____, so its velocity stays ____. That is called ____."

A CHECKS "So you're saying nothing pushes it forward. I agree / disagree because ____."

ROUND 2 · SWITCH · A CART: 4 N RIGHT, 4 N LEFT, STILL MOVING



B ASKS "The forces are balanced. Why is the cart still moving?"

A SAYS "Balanced means the net force is ____. The cart keeps a ____ velocity."

B ADDS "I want to add that balanced does not mean ____."

ROUND 3 · BOTH · A 1 kg CART vs. A 3 kg CART



A "The ____ kg cart is harder to stop because it has more ____."

B "More mass means more inertia, so ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Things keep moving because…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R6 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Three inertia myths are on the loose

09 Y

Fill each poster: the truth and evidence that proves it.

WANTED

FOR CRIMES AGAINST PHYSICS

"Moving things need a force to keep moving."

LAST SEEN: a puck sliding on ice

THE TRUTH:

EVIDENCE:

REWARD: NEWTON'S 1st LAW

WANTED

FOR CRIMES AGAINST PHYSICS

"Balanced forces mean nothing moves."

LAST SEEN: a car on cruise control

THE TRUTH:

EVIDENCE:

REWARD: CONSTANT VELOCITY

WANTED

FOR CRIMES AGAINST PHYSICS

"Heavier objects have less inertia."

LAST SEEN: pushing a full vs. empty cart

THE TRUTH:

EVIDENCE:

REWARD: MORE MASS

CAPTURED! · WRITE YOUR OWN WANTED NOTICE

A wrong idea about today's physics I once believed:

R6 1 2 3

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The passenger, in six panels

09 E

DIRECTIONS · Draw the passenger in each panel. Add a velocity arrow and any force arrows. Fill each caption.

1 · CRUISING

Car + passenger at 15 m/s. Net force on passenger: _____

2 · BRAKES!

The car slows. The passenger keeps moving _____ .

3 · SEATBELT

The belt pushes the passenger _____ .

4 · STOPPED

Both at 0 m/s. Net force: _____

5 · GREEN LIGHT

The car speeds up. Passenger feels pushed _____ .

6 · THE TRUTH

Nothing pushed them back — their _____ resisted the change.

THE WHOLE STORY · v–t GRAPH FOR THE PASSENGER

WHY DOES 2026-XJ KEEP MOVING?

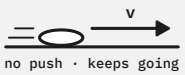
Use the words inertia, net force, and velocity.

R6 1 2 3

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Picture dictionary: inertia

Language goal: I can **explain** why things keep moving.



HARD TO CHANGE

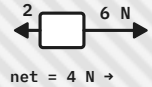
inertia

inercia · inércia

resisting a change in motion

my language: _____

use it: _____



net = 4 N →

→ + ← = 0

net force

fuerza neta · força resultante

all forces added up

my language: _____

use it: _____



net = 0

← 4 N 4 N →

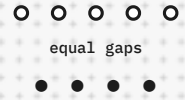
balanced

equilibrado · equilibrado

forces cancel: net = 0

my language: _____

use it: _____



constant velocity

velocidad constante · velocidade constante

same speed, same direction

my language: _____

use it: _____



kg

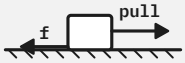
mass

masa · massa

how much matter + inertia

my language: _____

use it: _____



→ ← rub

friction

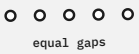
fricción · atrito

a force that slows sliding

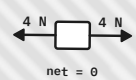
my language: _____

use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE



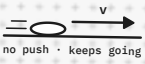
_____



_____



_____



_____

I can point and say all 6 words to a partner ☐

R6 (1) (2) (3)

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Fix the group chat

09 R

DIRECTIONS · Reply in the empty bubbles. Show your reasoning on the right.

XJ-7 tracking crew

ELI

engines off. so 2026-XJ will start slowing down now, right?

YOU ①

MAZ

but what keeps pushing it forward?

YOU ②

NADIA

test cart: pull 4 N right, friction 4 N left. it's moving. how??

YOU ③

ELI

ok which is harder to stop, a 1 kg or 3 kg cart at the same speed?

YOU ④

SHOW YOUR RECEIPTS

① 1st law: ΣF = 0 →

② no "force of motion"

③ balanced ≠ stopped

④ mass and inertia

R6 1 2 3

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Settle the argument: inertia

09 I

DIRECTIONS · Shade the bubble of the person who is right. Give evidence, then fix the wrong claim.

ROUND 1

ANA
"The puck slides because the push is still inside it."

BEN
"The push ended. With no net force, it keeps its velocity."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"Balanced forces mean the object must be stopped."

LU
"Balanced forces mean it's stopped OR moving at constant velocity."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"Real carts slow down because motion naturally runs out."

RAJ
"Friction and air are forces. They slow the cart."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

R6 1 2 3

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Comic: no push needed

09 CS

YOUR READER
A 3rd grader who thinks things need a push to keep moving.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **PURPLE** Same-length velocity arrows in panels 1 and 2
- ☐ ◌ **ORANGE** Force arrows only in panel 3, where a force acts
- ☐ ◯ **PENCIL** "net force = 0" written in panel 2

1 · THE SLIDE *A hockey puck sliding across smooth ice.*

CAPTION _____

2 · STILL SLIDING *Nobody touches it. It keeps going.*

CAPTION _____

3 · WHAT STOPS IT *It finally hits something rough or a wall.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **inertia · net force · velocity · friction**

R6 1 2 3

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Doodle Notes · inertia

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Inertia · inercia (ES) · inércia (PT)



TODAY'S CASE

A puck glides right on nearly frictionless ice after the hand releases it.

01 · SKETCH AND LABEL

Sketch the released puck at three equal time intervals. Draw purple velocity arrows for zero net force.

02 · CHANGE THE PICTURE

Now a brief leftward push acts. Draw a second sequence that shows what changes during that push. Label the force in orange.

Annotate your sketches. Circle the mark that best shows inertia. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

After release the puck keeps moving because ...

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · inertia

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Inertia · inercia (ES) · inércia (PT)

A · Inertia is an extra forward push acting on the cart.

B · A released cart keeps its velocity when the net force is zero.

01 · DEFINITION

What does inertia mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else